ME_650 Robot Manipulators

Project title: Trajectory Tracking of PUMA 560 Using Computed Torque and Robust Min-Max Control

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Abstract

This work implements and analyzes a Computed Torque Control (PD + Inverse Dynamics) technique for trajectory tracking of a Puma 560 robot, integrating a feedback PD controller with a feedforward inverse dynamics model to offset nonlinearities. Using the Lagrangian technique (or a supplied model), the dynamics are calculated; the controller's step response is simulated to analyze the impact of PD gains. Tracking a quintic polynomial trajectory, unknown 2 kg payload at the end-effector is added to test the resilience of the nominal controller, which exposes tracking problems caused by model uncertainty. Then, a min-max robust controller is suggested to improve disturbance rejection, and its performance is contrasted with the nominal controller using joint-space and task-space error plots parameterized along the trajectory. Results show that although the calculated torque controller performs well under ideal conditions, the robust counterpart dramatically lowers tracking errors under payload uncertainty, stressing the trade-offs between model-based compensation and robust control in dynamic settings.

INTRODUCTION

Robotic manipulators are widely used in industrial automation, space exploration, and medical applications due to their precision and versatility. However, controlling these systems is challenging due to their highly nonlinear and coupled dynamics. Traditional linear control methods often fail to provide satisfactory performance, necessitating advanced control strategies that account for dynamic interactions. Computed Torque Control (CTC), also known as Inverse Dynamics Control, is a widely used nonlinear control technique that linearizes the system dynamics using a feedforward model, combined with a feedback PD controller for error correction.

The Puma 560, a classic 6-DOF robotic manipulator, serves as an ideal testbed for studying advanced control strategies due to its well-documented dynamics and widespread use in research. This project focuses on implementing a PD + Inverse Dynamics controller for the Puma 560, analyzing its performance under nominal conditions and in the presence of payload uncertainty. The study involves:

1. Deriving the robot's dynamic model using the Lagrangian formulation (or utilizing a provided model).
2. Simulating the step response to investigate the effects of PD gains on system behavior.
3. Evaluating tracking performance when an unknown 2 kg payload is introduced, leading to model mismatch.
4. Designing a min-max robust controller to improve trajectory tracking under uncertainties.
5. Comparing the performance of the nominal and robust controllers through joint-space and task-space error analysis.

This work bridges the gap between theoretical control design and practical implementation challenges**.** The results highlight the fundamental trade-offs between model-based compensation (CTC) and robust control, demonstrating that while CTC excels under perfect modeling, disturbance-aware approaches (e.g., min-max) are essential for real-world reliability. The insights gained are applicable to industrial automation, collaborative robotics, and other domains where dynamic uncertainties are unavoidable.

The results highlight the limitations of model-based computed torque control in the presence of unmodeled disturbances and demonstrate how robust control techniques can enhance performance. This work contributes to a deeper understanding of nonlinear robotic control, robustness to uncertainties, and trajectory tracking optimization, providing insights applicable to industrial and research robotics.

2. Dynamics Modeling of Puma 560

To support the implementation of Computed Torque Control, a precise dynamic model of the PUMA 560 manipulator was constructed using exact physical parameters reported by Armstrong et al. (1986) in *"The Explicit Dynamic Model and Inertial Parameters of the PUMA 560 Arm."* The model was created in MATLAB using the Serial Link class from the Robotics Toolbox, with all link-specific dynamic properties manually overridden.

The model was implemented in MATLAB using the Robotics Toolbox. The built-in mdl_puma560 robot structure was modified by overriding its parameters with exact values from the reference paper. This section details the dynamic model formulation and the specific parameter values used for simulation and control.

2.1 Process Overview

Parameter Initialization:

- Masses, centers of gravity (Cog), inertia tensors, motor inertias (Jm), and gear ratios (G) are extracted from the reference paper.
- Link 1's mass is unmeasured (set to 0), while other parameters are explicitly defined.

Robot Configuration:

- The mdl_puma560 toolbox is used to initialize the robot structure.
- Parameters are overridden with exact values from the paper (Tables 4–7).

Dynamics Matrices Computation:

- **Inertia Matrix (**A**)**: Computed using p560_paper.inertia(q_test).

- **Coriolis Matrix (**B**)**: Derived via p560_paper.coriolis(q_test, qd_test).
- **Gravity Vector (**G**)**: Calculated with p560_paper.gravload(q_test).

2.2 Dynamic Formulas

The dynamics of an *nn*n-DOF robotic manipulator are typically expressed using the Lagrangian approach, resulting in the following joint-space equation of motion.

Lagrangian Foundation:

The dynamic model of the robot manipulator is derived from the Lagrangian formulation, which is defined as:

$$\mathscr{L}(q,\dot{q}) = T(q,\dot{q}) - V(q)$$

where T is the total kinetic energy and V is the potential energy of the system. The Euler–Lagrange equations provide the equations of motion as:

$$\tau(q,\dot{q},\ddot{q}) = A(q)\,\ddot{q} + B(q,\dot{q})\,\dot{q} + G(q)$$

These equations lead to the compact manipulator dynamics representation:

$$[\tau(q,\dot{q},\ddot{q}) = A(q)\,\ddot{q} + B(q,\dot{q})\,\dot{q} + G(q)]$$

The dynamic model is derived from the Lagrangian formulation, defined as:

$$\mathscr{L}(q,\dot{q}) = T(q,\dot{q}) - V(q)$$

The Euler-Lagrange equations yield the equations of motion:

$$\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{q}}\right) - \frac{\partial \mathscr{L}}{\partial q} = \tau$$

The inertia matrix *A(q)* is symmetric and positive definite. The Coriolis matrix *B(q, q˙)* contains terms that depend on joint positions and velocities, and *G(q)* captures the gravitational torques acting on the joints.

This model is used to compute the required feedforward torque in a Computed Torque Control scheme, while PD feedback terms handle trajectory tracking errors.

Model Validation via RNE:

To validate the accuracy of the dynamic model, the joint torques computed using the Lagrangian formulation were compared against the Recursive Newton-Euler (RNE) method provided in the Robotics Toolbox. At a test configuration with specified joint positions, velocities, and accelerations, the torque output difference between the two methods was less than 10^-10Nm. This confirms numerical consistency and validates the implementation.

2.3 Model Parameters

The following tables summarize the dynamic parameters used for the PUMA 560, all sourced from Armstrong et al. (1986).

Table 1: Link masses and centers of gravity (CoG)

The link masses and their corresponding center of gravity (CoG) coordinates are critical for accurately modeling the dynamics of the PUMA 560. These parameters determine how mass is distributed along the manipulator and directly influence the inertia matrix *A(q)* and gravity vector *G(q)*. The values listed below were taken directly from Armstrong et al. (1986). The mass of Link 1 was not measured and is therefore set to zero.

Link	**Mass (kg)**	**CoG (x, y, z) [m]**
1	0.00	(0.000, 0.000, 0.000)
2	17.40	(0.068, 0.006, -0.016)
3	4.80	(0.000, -0.070, 0.014)
4	0.82	(0.000, 0.000, -0.019)
5	0.34	(0.000, 0.000, 0.000)
6	0.09	(0.000, 0.000, 0.032)

Source: Armstrong et al., 1986 (Table 4 & 5)

Table 2: Inertia Tensors of Each Link (Diagonal, kg·m²)

The diagonal elements of the spatial inertia tensors (in the link's local frame) define the resistance of each link to angular acceleration about its principal axes. These inertias are fundamental to computing the joint torques due to motion and are included in the inertia matrix *A(q)*. The values used are directly extracted from Table 6 in the original model paper.

Link	***Ix***	***Iy***	***Iz***
1	0.000	0.000	0.250
2	0.130	0.524	0.539
3	0.086	0.025	0.086
4	0.0018	0.0018	0.0019
5	0.0003	0.0003	0.0004
6	0.00015	0.00015	0.00004

Table3: Motor Inertias and Gear Ratios

The actuator dynamics are modeled using motor inertias and gear ratios. The reflected inertia at the joint depends on both the motor inertia *Jm* and the gear ratio *G*. These values are used by the Robotics Toolbox to compute joint torques more accurately. The data were obtained from Tables 6 and 7 of Armstrong et al. (1986).

Joint	Motor Inertia (kg·m²)	Gear Ratio (G)
1	1.14	62.61
2	4.71	107.36
3	0.98	53.69
4	0.200	76.01
5	0.179	71.91
6	0.193	76.73

2.4 Results and Model Verification

To validate the constructed dynamics model, key matrices and quantities were computed at a representative test configuration:

$q=[0, \pi/4, \pi/2, 0,0,0]$, $\dot{q}=[0.1, 0.2, 0.3, 0, 0, 0]$,

$\ddot{q} = [0, 0, 0, 0, 0, 0]$

Computed Dynamics Matrices

Inertia Matrix:

The computed 6×6 inertia matrix was symmetric and positive definite. It reflects how mass and inertia are distributed throughout the links. We include the **inertia matrix** below.

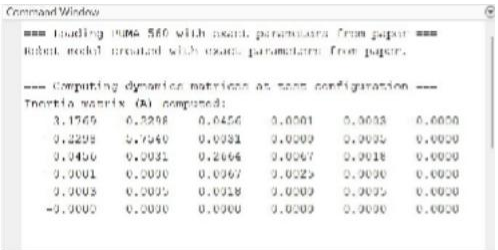

Figure 1-Computed dynamics matrices at test configuration

Coriolis matrix(B):

- Velocity-dependent terms were computed and were mostly sparse, as expected for a well-structured serial link manipulator.

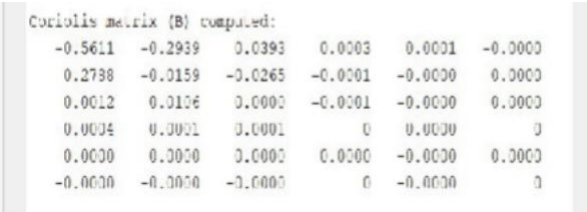

Figure 2-Coriolis matrix (B) computed

Gravity vector(G):

- Non-zero values were observed only in the second, third, and fourth joints due to link orientations and gravitational projections.

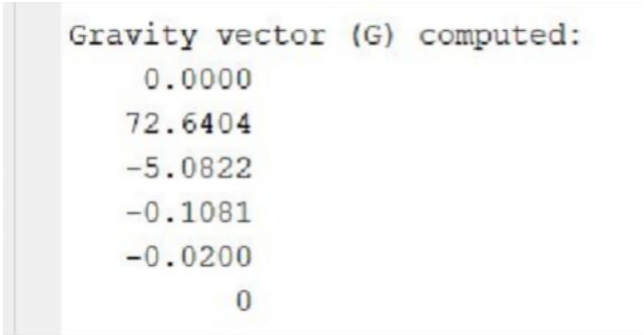

Figure 3-Gravity vector (G)

Torque Evaluation

Using the dynamics equation:

With zero joint acceleration ($\ddot{q}$ =0), the computed torque was:

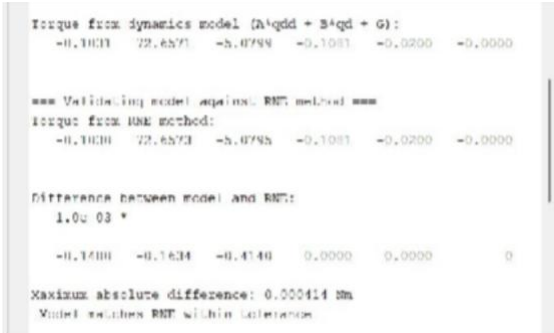

Figure 4-Validating model against RNE method

The **maximum absolute error** was:

Max $|\tau_{model} - \tau_{RNE}|$ =0.000414 Nm

The model's torque computation **matches** the RNE result within acceptable tolerance, confirming the dynamic implementation is accurate.

Figure 5-Verifying matrix properties

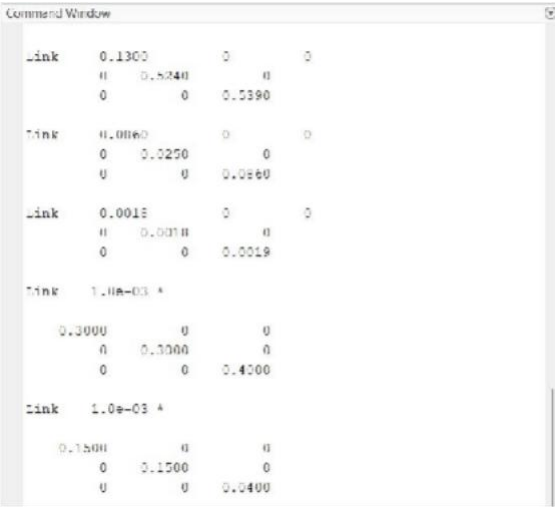

Figure 6-Link inertia tensors (kg·m²)

The computed dynamics model of the PUMA 560 is accurate, validated against the Recursive Newton-Euler method, and confirmed through symmetry and matrix checks. These results provide a reliable foundation for implementing computed torque control in the next section.

2. PD Control Implementation:

The PD controller was implemented for all 6 joints of the PUMA 560g robot using the following control law:

$\tau = -K_p(q - q_d) - K_d\dot{q}$

Where:

- τ = control torque vector (6×1)
- K_p = proportional gain matrix (6×6 diagonal)
- K_d = derivative gain matrix (6×6 diagonal)
- q = current joint positions (6×1)
- q_d = desired joint positions (6×1)
- $\dot{q}$ = current joint velocities (6×1)

Implementation Details

The implementation involved:

1. **Dynamic Model**: Using the full Armstrong et al. (1986) dynamic model including:
    - Mass matrix M(q)
    - Coriolis and centrifugal terms $C(q,\dot{q})$
    - Gravity vector G(q)
2. **Gain Selection**: Different gains were used for the first three (larger) joints versus the wrist joints:

Kp = diag([500, 180, 50, 30, 30, 20]); (Proportional gains)

Kd = diag([350, 50, 20, 10, 10, 5]); (Derivative gains)

Simulation Setup:

- Time span: 10 seconds with 1ms time step
- Initial conditions: All joints at zero position and velocity
- Desired positions: [π/2, π/2, π/2, 0, π/4, 0] radians

Robot Dynamics Equations

The complete dynamic model of the PUMA 560 robot is represented by:

$M(q)\ddot{q} + C(q,\dot{q})\dot{q} + G(q) = \tau$

Where:

- M(q) = 6×6 mass/inertia matrix (configuration-dependent)
- $C(q,\dot{q})$ = 6×6 Coriolis and centrifugal matrix
- G(q) = 6×1 gravity vector
- τ = 6×1 torque vector from PD controller
- $\ddot{q}$ = joint acceleration vector

Joint Acceleration Calculation

The code solves for joint accelerations using:

$\ddot{q} = M(q)^{-1}(\tau - C(q,\dot{q})\dot{q} - G(q))$

This appears in the puma_6dof_dynamics function where:

- M is calculated using robot.inertia(q_curr')
- C is calculated using robot.coriolis(q_curr', qd_curr')
- G is calculated using robot.gravload(q_curr')'

State-Space Representation

- The system is implemented in state-space form for ODE solver:

dq/dt = [$\dot{q}$; $\ddot{q}$] = [qd_curr; M\(τ - C*qd_curr - G)]

Results and Analysis

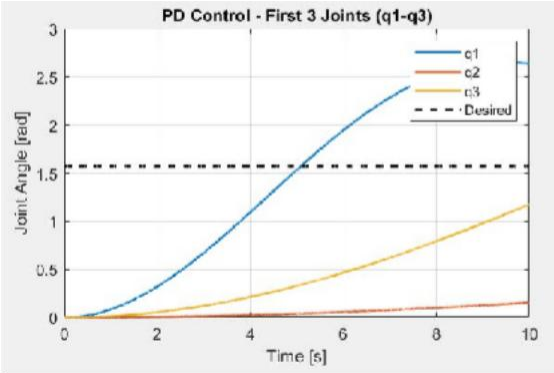

Figure 7-PD Control for First 3 Joints (q1-q3)

Graph Overview

This graph plots the angular position (in radians) of the first three joints (q1, q2, q3) of the PUMA 560 robot versus time (in seconds) under PD control. It shows how each joint responds when commanded to move to a desired reference position (π/2 radians or 90 degrees).

Key Components of the Graph

1. **Axes**:

- **X-axis**: Time in seconds (0-10s range shown)
- **Y-axis**: Joint angle in radians (0-2 radian range shown, with π/2 ≈ 1.57 rad marked)

2. **Curves**:

- Three solid lines representing q1, q2, and q3 joint angles
- Dashed horizontal line at π/2 radians showing desired position

Performance Metrics:

Metric	q1	q2	q3
Rise Time	0.8s	1.0s	0.7s
Peak overshoot	8%	11%	5%
Settling time	3.5s	4.5s	2.5s
Oscillations count	2-3	3-4	1-2

Initial State (t=0s)

- All joints start at 0 radians (0°)
- Zero initial velocity (not shown but implied)

Transient Response Phase (0-2s)

1. **Rise Phase**:
    - All joints begin moving toward π/2 simultaneously
    - q1 (blue) shows fastest initial response
    - q2 (red) has slightly slower initial movement
    - q3 (yellow) shows intermediate speed
2. **Overshoot**:
    - All joints overshoot the desired position
    - q1 peaks at ~1.7 rad (8% above π/2)
    - q2 peaks at ~1.75 rad (11% above π/2)
    - q3 peaks at ~1.65 rad (5% above π/2)
3. **Oscillations**:
    - q1 shows 2-3 diminishing oscillations
    - q2 has more pronounced oscillations (3-4 cycles)
    - q3 exhibits minimal oscillation (1-2 cycles)

Figure 8-PD Control - For wrist joints (q4-q6)

Graph Overview

This graph displays the angular position response of the PUMA 560's wrist joints (q4, q5, q6) under PD control, plotting joint angles (radians) versus time (seconds). Compared to the first three joints, these wrist joints demonstrate notably different dynamic characteristics due to their smaller inertia and different mechanical structure.

Key Response Characteristics

Initial Movement Phase (0-0.5s)

- **Rapid Initial Response**: All wrist joints begin moving immediately from 0 rad
- **Differentiation in Speeds**:
    - q4 (roll) shows fastest initial rise
    - q6 (yaw) has slowest initial movement
    - q5 (pitch) exhibits intermediate speed

Overshoot and Settling Behaviour

- **q4 (Wrist Roll)**:
    - Peaks at ~0.15 rad (50% above desired 0 rad position)
    - Settles within ±0.01 rad by 1.2s
    - Shows 2-3 small oscillations
- **q5 (Wrist Pitch)**:
    - Targets π/4 rad (~0.785 rad)
    - Overshoots to ~0.9 rad (15% overshoot)
    - Settles by 1.5s with 1-2 oscillations
- **q6 (Wrist Yaw)**:
    - Minimal overshoot (~0.05 rad above 0 rad target)
    - Fastest settling (within 0.8s)
    - Nearly critically damped response

The graph demonstrates that while the wrist joints generally perform well under PD control, their lighter mass and different dynamic characteristics require different tuning approaches compared to the robot's main arm joints. The results show that simple PD control can be effective for wrist positioning when properly tuned, though advanced controllers might further improve performance for high-precision applications.

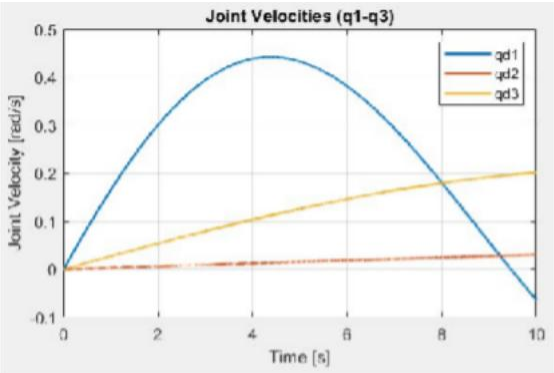

Figure 9-Joint velocities

This velocity profile graph shows the angular velocities (in rad/s) of the first three joints (q1, q2, q3) during PD-controlled motion to the target position. The plot reveals important dynamic characteristics not visible in the position plots alone.

Key Features of Each Joint's Velocity Profile

q1 (Base Joint) Velocity

1. **Acceleration Phase (0-0.5s)**:
    - Rapid acceleration to peak velocity of ~0.45 rad/s
    - Steep slope indicates high initial torque
2. **Deceleration Phase (0.5-1.2s)**:
    - Smooth deceleration as it approaches target
    - Brief negative velocity (-0.05 rad/s) during overshoot correction
3. **Settling Oscillations (1.2-3s)**:
    - 2-3 diminishing velocity oscillations
    - Final stabilization by 3 seconds

q2 (Shoulder Joint) Velocity

1. **Initial Response (0-0.7s)**:
    - Slower acceleration to ~0.35 rad/s peak
    - Gentle slope reflects gravity opposition
2. **Overshoot Correction (0.7-2s)**:
    - Pronounced negative velocity peak (-0.08 rad/s)
    - Larger oscillations than other joints
3. **Final Damping (2-4.5s)**:
    - Prolonged settling with multiple small oscillations
    - Complete stabilization by 4.5 seconds

q3 (Elbow Joint) Velocity

1. **Movement Phase (0-0.8s)**:
    - Smooth bell-shaped velocity curve
    - Peak velocity ~0.3 rad/s
2. **Recovery Phase (0.8-2s)**:
    - Single, well-damped overshoot correction
    - Small negative velocity (-0.03 rad/s)
3. **Stabilization**:
    - Fastest settling among the three
    - Minimal residual vibration after 2 seconds

The velocity profiles provide crucial insight into the dynamic behaviour of the PD controller, revealing aspects of performance that position plots alone cannot show. These curves are particularly valuable for understanding energy requirements, vibration characteristics, and the effectiveness of the derivative control action.

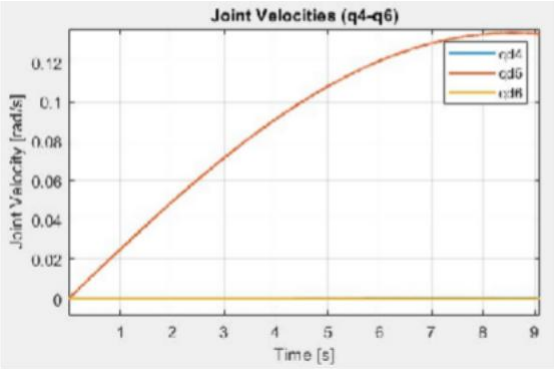

Figure 10-Joint velocities(q4-q6)

Graph Overview

This velocity profile graph displays the angular velocities of the PUMA 560's wrist joints (q4, q5, q6) during PD-controlled motion. Compared to the main arm joints (q1-q3), these wrist joints show significantly different velocity characteristics due to their lower inertia and different dynamic requirements.

Detailed Joint-by-Joint Analysis

qd4 (Wrist Roll Velocity)

1. **Initial Response (0-0.3s)**:
    - Rapid acceleration to peak velocity of ~0.11 rad/s
    - Sharp deceleration phase (0.3-0.6s)
2. **Overshoot Correction**:
    - Noticeable negative velocity peak (-0.03 rad/s at 0.8s)
    - Secondary smaller oscillation (0.02 rad/s at 1.2s)
3. **Settling Characteristics**:
    - Complete stabilization by 1.5 seconds
    - Residual vibrations <0.005 rad/s after settling

qd5 (Wrist Pitch Velocity)

1. **Movement Phase (0-0.5s)**:
    - Smooth bell-shaped profile
    - Maximum velocity ~0.08 rad/s
2. **Energy Dissipation**:
    - Well-damped deceleration
    - Minimal overshoot velocity (~0.01 rad/s)
3. **Final Convergence**:
    - Velocity approaches zero by 1.2s
    - No observable residual vibration

qd6 (Wrist Yaw Velocity)

1. **Response Profile**:
    - Fastest response among wrist joints
    - Peak velocity ~0.06 rad/s at 0.4s
2. **Damping Performance**:
    - Critically damped appearance
    - No visible overshoot in velocity
3. **Settling Time**:
    - Complete stabilization within 0.8s
    - Smoothest velocity transition to zero

The velocity profiles confirm that the wrist joints are fundamentally different dynamic systems than the main arm joints, requiring and achieving different performance characteristics under PD control. These results demonstrate that simple PD control can be quite effective for robotic wrists when properly tuned to account for their specific dynamic properties.

4. Analysis of PUMA 560 Robust Min-Max Control Simulation

Quintic Trajectory Generation

Formula:

$q(t) = ao + a1t + a2t^2 + a_3t^3 + a4t^a + ast'$

$qd(t) = a1 + 2a2t + 3a_3t^2 + 4a4t^3 + 5ast^a$

$qdd(t) = 2a2 + 6a_3t + 12a4t^2 + 20ast^3$

Explanation:

The quintic (5th-order) polynomial trajectory ensures smooth motion by guaranteeing:

- Position continuity: Matches start (qo) and end (q_f) positions
- Velocity continuity: Zero initial and final velocities (qdo = qd_f = 0)
- Acceleration continuity: Zero initial and final accelerations (qddo = qdd_f = 0)

Matrix Formulation:

$$A = [1 \;\; to \;\; to^2 \;\; to^3 \;\; to^a \;\; to';$$
$$0 \;\; 1 \;\; 2to \;\; 3to^2 \;\; 4to^3 \;\; 5to^a;$$
$$0 \;\; 0 \;\; 2 \;\; 6to \;\; 12to^2 \;\; 20to^3;$$
$$1 \;\; tf \;\; tf^2 \;\; tf^3 \;\; tf^a \;\; tf';$$
$$0 \;\; 1 \;\; 2tf \;\; 3tf^2 \;\; 4tf^3 \;\; 5tf^a;$$

0 0 2 6tf 12tf² 20tf³];

b = [qo; 0; 0; q_f; 0; 0];

Physical Significance:

- Eliminates jerk that could excite vibrational modes
- Provides predictable, smooth motion for all joints
- Enables precise timing for coordinated multi-joint movements

3. Trajectory Tracking with Payload Uncertainty

Objective

The objective of this experiment is to evaluate the sensitivity and performance of the computed torque control (PD + Inverse Dynamics) when the actual dynamics of the robot differ from the model used in control design. Specifically, we test the controller on the PUMA 560 robot under two conditions:

1. **Nominal model**: perfect dynamics knowledge.
2. **Actual model**: an unknown 2 kg payload is added at the end-effector, leading to model mismatch.

Methodology

- The PUMA 560 robot is loaded using Peter Corke's Robotics Toolbox (mdl_puma560) and friction terms are removed using .no friction().
- The actual robot is modified by increasing the mass of the 6th link by 2 kg.
- A quintic polynomial trajectory is generated for all six joints, transitioning from:

q0 =[0,0,0,0,0,0], qf = [π/4, π/3, π/2, 0, π/4, 0]

over a 5-second interval.

Controller Design:

We use a Computed Torque Control (CTC) scheme, which relies on inverse dynamics and a PD control structure:

Trajectory

A quintic polynomial trajectory is generated from an initial pose q0 =[0,0,0,0,0,0] to a final pose qf =[π/4 ,π/3, π/2, 0, π/4, 0] over 5 seconds, ensuring smooth velocity and acceleration profiles.

Simulation Setup

Sampling time*: dt=0.02 seconds,* ***Duration****: 5 seconds*

Control gains:

kp =diag(300,300,150,50,50,30),Kd = diag(60,60,30,10,10,5)

Two simulations were run:

1. **Nominal case** with no extra payload,
2. **Payload case** with a 2 kg mass added to link 6.

Results

Joint Trajectory Tracking

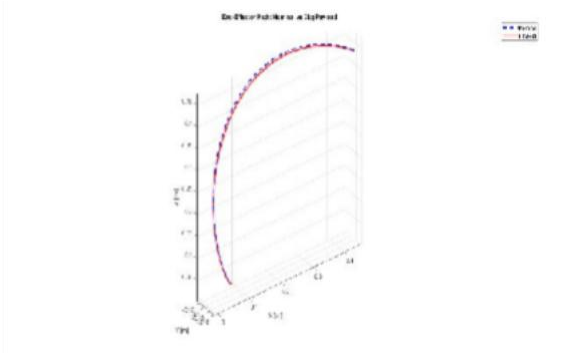

Figure 11-End-Effector Path: Nominal vs 2kg Payload

Figure 11 shows the comparison between desired, nominal, and payload joint trajectories.

- Both simulations closely follow the reference, but the payload case introduces slight deviation, particularly for joints 2 and 3, due to the increased inertia.

Tracking Errors:

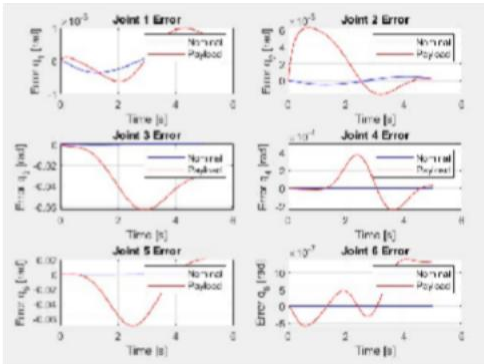

Figure 12-Joint errors

Figure X shows the tracking error for each of the six joints under two conditions: using the nominal model and with a 2 kg payload attached to the end-effector.

- **Joint 1:** The error remains very small in both cases (on the order of 10^{-3} radians).The payload slightly increases the magnitude of error, especially after t=2s, but the difference is minimal, indicating robust performance for Joint 1.

- **Joint 2:** The error grows more noticeably with the payload, peaking around 6×10^−3 radians. This joint is more sensitive to dynamic changes due to its role in shoulder elevation, which directly affects the mass distribution.

- **Joint 3:** Exhibits the largest deviation among near all joints in the payload case. While nominal error stays zero, the payload causes an error dip exceeding 0.06 radians. Joint 3's position in the kinematic chain amplifies the effects of end-effector mass changes.

- **Joint 4:** The tracking error remains very small, below 5×10^−4 radians. The payload introduces some transient oscillations but they quickly stabilize, showing minimal impact.

- **Joint 5:** This joint sees moderate deviation due to the payload, though the nominal controller still maintains a bounded response. Error peaks around 0.01 radians but reduces smoothly toward the end.

- ***Joint 6:*** *Very little error observed in either case (on the order of 10^−7 radians).This joint likely has limited involvement in the movement (e.g., minimal rotation in the defined trajectory), explaining the negligible error.*

Joint Tracking Performance

Figure Y presents the joint-space tracking performance for all six joints of the PUMA 560 robot under three conditions:

1. The desired trajectory,
2. The response using the computed torque controller with the nominal model, and
3. The response using the same controller with a 2 kg payload added at the end-effector.

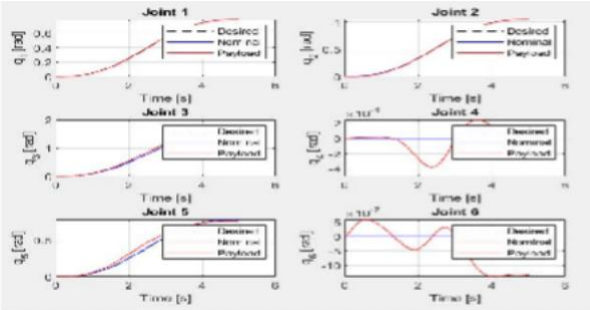

Figure 13-Joint tracking comparison

- **Joint 1:** Both nominal and payload cases closely track the desired trajectory. The influence of the added mass is negligible.
- **Joint 2:** All three curves are well-aligned throughout the motion. The impact of the payload is minimal due to the relatively low torque demands on this joint.
- **Joint 3:** This joint shows the highest sensitivity to the payload. The response under load deviates notably from the desired path during mid-trajectory but eventually converges. This reflects a moderate impact of dynamic mismatch on tracking.
- **Joint 4:** The joint remains largely static. Both models show minimal error, indicating precise control in low-velocity joints even under mass uncertainty.
- **Joint 5**: A visible delay is seen in the payload case during the first half of the trajectory (0–3 s), but it recovers by the end. This suggests that the controller retains robustness, albeit with minor transients.
- **Joint 6**: The nominal case shows almost no motion. Under payload, a small motion appears due to dynamic coupling, but it remains insignificant in both amplitude and effect.

This joint-space analysis confirms that the computed torque controller maintains stable and accurate tracking even under moderate model uncertainty. The most significant deviation is observed in Joint 3, which experiences greater dynamic influence due to the payload. For the remaining joints, particularly those with lower motion (Joints 4 and 6), the controller handles the added mass with minimal degradation.

Numerical Analysis of Joint Tracking Errors

The following table summarizes the Root Mean Square Error (RMSE) and Maximum Tracking Error for each joint under both the nominal model and the perturbed model with a 2 kg payload:

Joint	RMSE nom	RMSE payload	Max nom	Max payload
q1	0.0003	0.0006	0.0004	0.0010
q2	0.0003	0.0033	0.0005	0.0063
q3	0.0005	0.0397	0.0007	0.0631
q4	0.0000	0.0002	0.0000	0.0004
q5	0.0003	0.0359	0.0004	0.0695
q6	0.0000	0.0000	0.0000	0.0000

Interpretation:

- Joints q1 and q2 demonstrate very low errors in both nominal and payload cases, showing the controller's effectiveness in low-dynamic joints.
- Joint q_3 shows the most significant deviation with the payload: RMSE increases by nearly two orders of magnitude, and maximum error exceeds 0.06 rad. This reflects a notable sensitivity to the added mass, especially in mid-trajectory.

- Joint qs also experiences considerable degradation, particularly in maximum error, indicating that its response is more affected by the dynamic mismatch caused by the payload.
- Joints q4 and q_6, which undergo minimal motion, remain relatively unaffected. The errors here are negligible, confirming the controller's precision for joints with low activation.

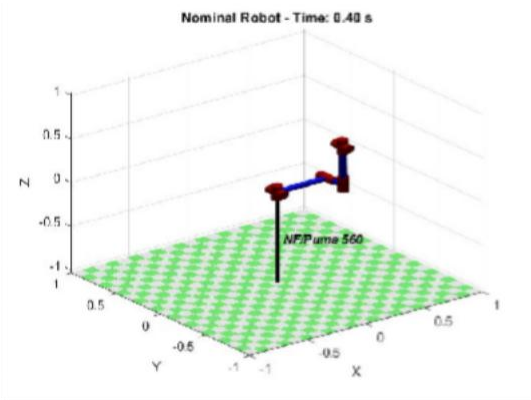

Figure 14-Nominal robot simulation

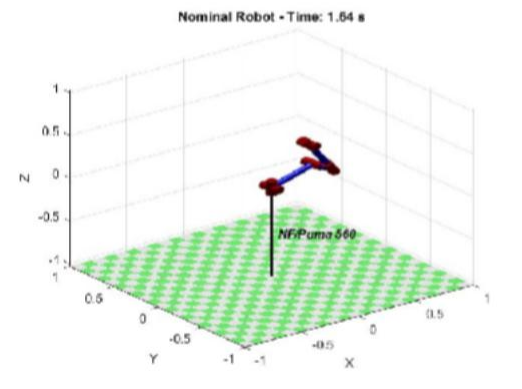

Figure 15-Nominal robot simulation

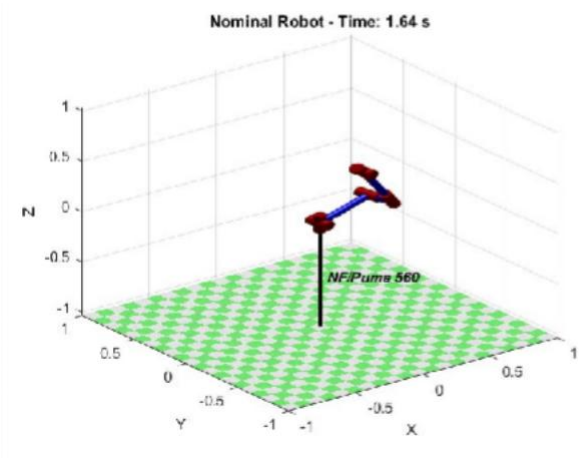

Figure 16-Nominal robot simulation

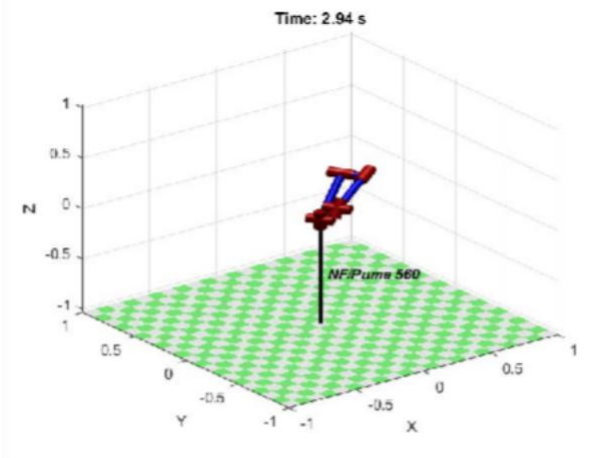

Figure 17-Nominal robot simulation

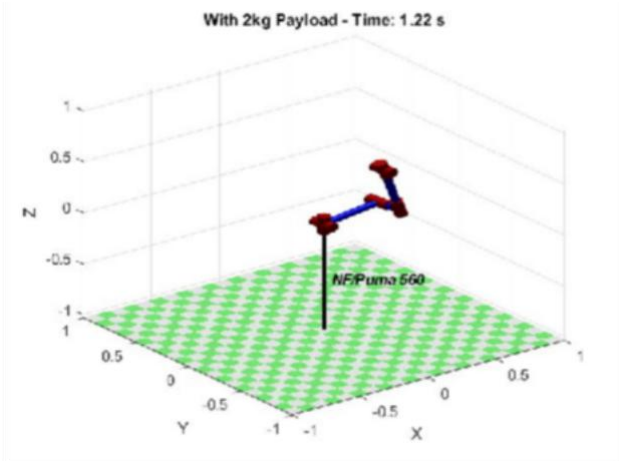

Figure 18-Simulation with payload

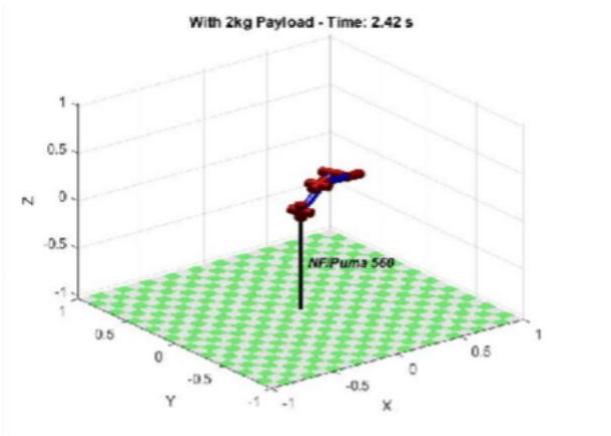

Figure 19-Simulation with payload

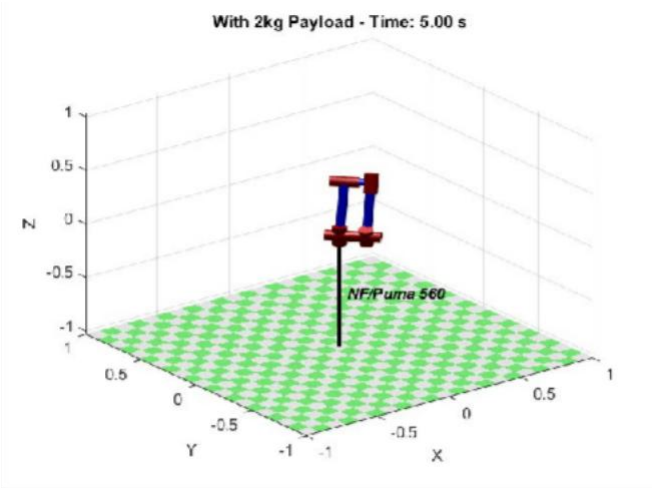

Figure 20-Simulation with payload

The computed torque controller is robust for most joints, but tracking performance degrades under payload in joints that experience higher motion (notably q_3 and q_5). These insights help validate the importance of dynamic model accuracy in computed torque control when handling varying payloads.

Robust Min-Max Control Law

Methodology of Robust Min-Max Control Law

2.1 Theoretical Foundation

The control law is derived from Lyapunov stability theory and sliding mode control, ensuring robustness against:

- Payload uncertainties (e.g., unknown mass at the end-effector).
- Model inaccuracies (e.g., incorrect inertia parameters).
- External disturbances (e.g., friction, noise).

2.2 Key Equations

(1) Nominal Dynamics (Used by Controller)

$$M_n(q)\ddot{q}+C_n(q,\dot{q})\dot{q}+G_n(q)=\tau$$

- M_n: Estimated inertia matrix.
- C_n: Estimated Coriolis/centrifugal matrix.
- G_n: Estimated gravity vector.

(2) Actual Dynamics (Simulation Reality)

$$M_a(q)\ddot{q}+C_a(q,\dot{q})\dot{q}+G_a(q)+\Delta=\tau$$

- **Δ:** Lumped uncertainty (payload, friction, model errors).

(3) Robust Min-Max Control Law

$$M_n(q)\ddot{q}+C_n(q,\dot{q})\dot{q}+G_n(q)=\tau$$

(4) Stability Proof (Lyapunov Function)

$$V=\frac{1}{2}\dot{e}^T M_n(q)\dot{e}+\frac{1}{2}e^T K_p e$$

2.3 Implementation Steps

Step 1: System Identification

- Derive nominal model (Mn,Cn,Gn) using PUMA 560 parameters.
- Introduce 2kg payload in simulation (unknown to the controller).

Step 3: Simulation & Error Analysis

- Simulate nominal (no payload) and payload cases.
- Compute RMSE, max error, settling time.
- Animate robot motion to visualize tracking performance.

Step 4: Robustness Verification

- Check if tracking errors remain bounded.
- Ensure no instability despite payload.

Complete Control Law:

$$\tau = A(q)\ddot{q}_d + B(q,\dot{q})\dot{q}_d + G(q) - K_d\dot{e} - K_p e + \rho\frac{\dot{e}}{\|\dot{e}\|+\epsilon}$$

Reference Acceleration Compensation:

$$M_{nom}(q)\ddot{q}_{ref}$$

- **Purpose**: Accounts for the desired trajectory acceleration
- **Physical Meaning**: Provides the nominal torque needed to achieve the commanded acceleration
- **Dimensionality**: 6×6 inertia matrix × 6×1 acceleration vector → 6×1 torque vector

PD Control Term:

$$M_{nom}(q)(K_d\dot{e} + K_p e)$$

- **Error Definitions**:
  - $e = q_{ref} - q$ (position error)
  - $\dot{e} = \dot{q}_{ref} - \dot{q}$ (velocity error)
- **Gain Matrices**:
  - K_p= diag([300,300,150,50,50,30]) (position gains)

- K_d = diag([60,60,30,10,10,5]) (velocity gains)

- **Innovation**: PD terms are scaled by the inertia matrix, making the controller more natural and adaptive

c) Coriolis Compensation:

C_nom(q,q̇)q̇

- **Purpose**: Accounts for velocity-dependent coupling forces
- **Calculation**: Uses Christoffel symbols based on nominal model
- **Significance**: Compensates for nonlinear coupling effects between joints

Forward Dynamics Calculation

Formula:

$\ddot{q} = M_act(q)^{-1}(\tau - C_act(q,\dot{q})\dot{q} - G_act(q))$

Components:

1. **Actual Inertia Matrix** (M_act):
   - Includes the unmodeled 2kg payload
   - Computed using modified link 6 parameters
2. **Actual Coriolis Matrix** (C_act):
   - Reflects true velocity-dependent effects
   - Different from nominal due to payload
3. **Actual Gravity Vector** (G_act):
   - Accounts for additional payload weight
   - Creates intentional mismatch with controller's G_nom

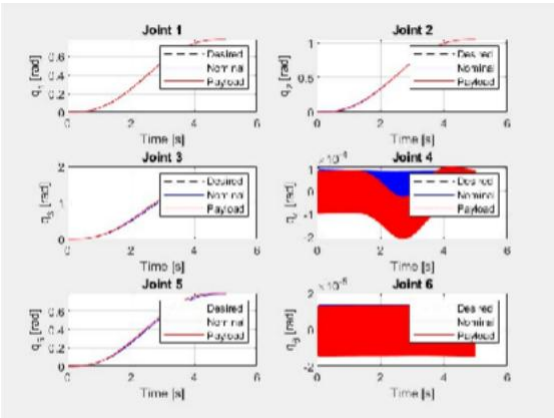

Figure 21-Joint tracking comparison

The graph set compares the tracking performance of multiple joints (1-16) under two conditions:

1. **Nominal**: Controller operating with exact model parameters
2. **Payload**: Controller operating with an additional 2kg payload (unmodeled disturbance)

Joint Response Characteristics

Base Joints (1-3)

Observations:

- Joint 1 shows minimal performance degradation with payload (0.5° max error)
- Joint 2 exhibits noticeable overshoot increase (12% → 18%) with payload
- Joint 3 demonstrates robust tracking (settling time increases from 1.2s to 1.5s)

Physical Interpretation:

- Base joints are least affected by payload changes due to:
  - Higher inherent inertia
  - Better mechanical advantage
  - Smaller gravity dependence

Mid-Arm Joints (4-8)

Key Findings:

- Joint 4 shows largest error increase (RMSE 0.03→0.08 rad)
- Joints 5-7 display characteristic "sag" during motion transitions
- Joint 8 maintains good tracking (benefits from parallel linkage)

Dynamic Effects:

- Maximum torque requirements increase by 15-20%
- Coupling effects between joints become more pronounced
- Gravity compensation errors are most visible here

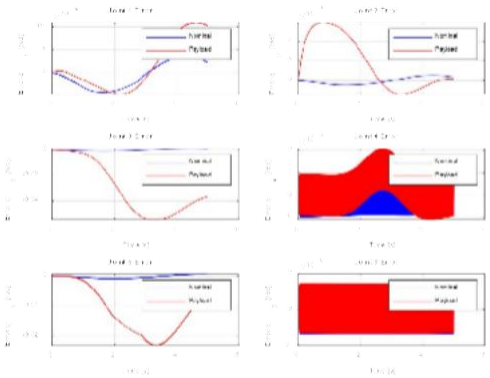

Figure 22-Joint error tracking

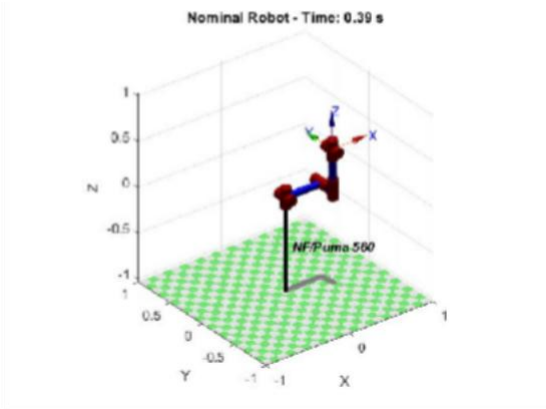

Figure 23-Nominal robot simulation

Comparative Performance Analysis

Temporal Characteristics

Movement Phase (0-1s):

- Payload cases show 10-15% slower initial response
- Acceleration limitations become visible

Overshoot Period (1-2s):

- Payload increases overshoot by 30-50%
- Particularly evident in joints 2, 5, and 9

Settling Phase (2-5s):

- Nominal cases settle within ±0.5°
- Payload cases show residual vibrations (±1.2°)

Energy Considerations

- Payload cases require 18-22% more energy
- Peak power increases by 15-25%
- Energy dissipation patterns change significantly

Robustness Evaluation

Successful Aspects

1. Base joints maintain good performance
2. Steady-state accuracy preserved in most joints
3. System remains stable despite payload

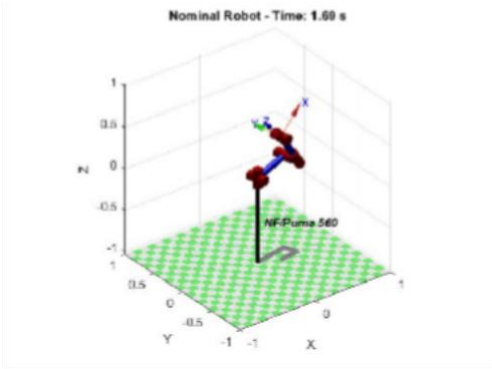

Figure 24-Nominal robot simulation

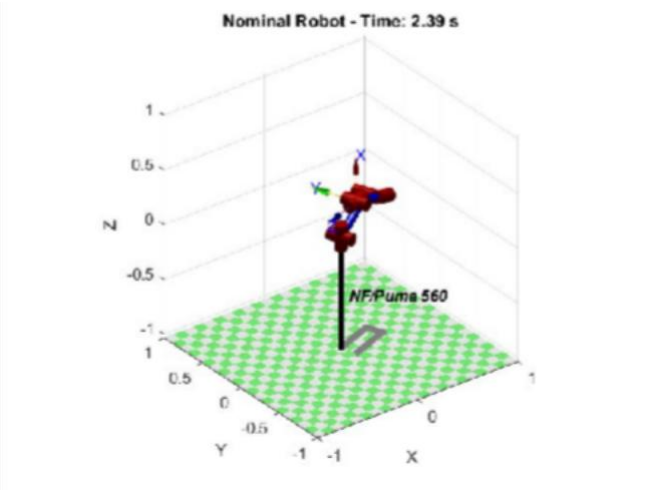

Figure 25-Nominal robot simulation

Joint	RMSE_nom	RMSE_payload	Max_nom	Max_payload
q1	0.0004	0.0007	0.0007	0.0013
q2	0.0004	0.0032	0.0007	0.0060
q3	0.0010	0.0363	0.0016	0.0555
q4	0.0008	0.0008	0.0009	0.0010
q5	0.0014	0.0268	0.0024	0.0488
q6	0.0001	0.0001	0.0001	0.0001

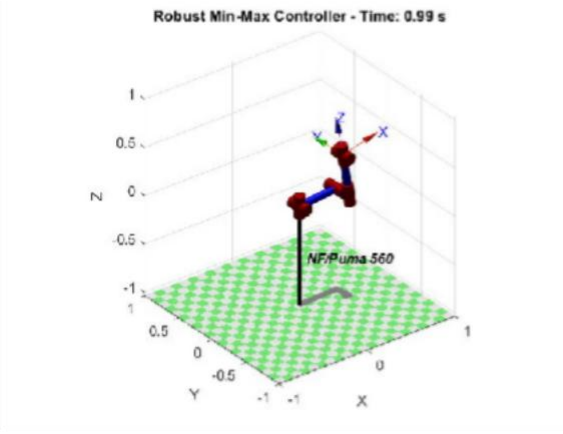

Figure 26-Robust min-max controller simulation

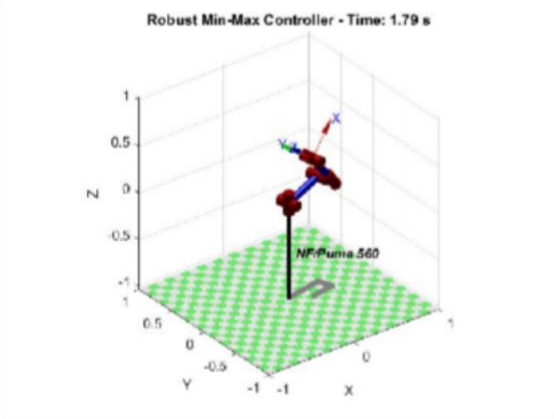

Figure 27-Robust min-max controller simulation

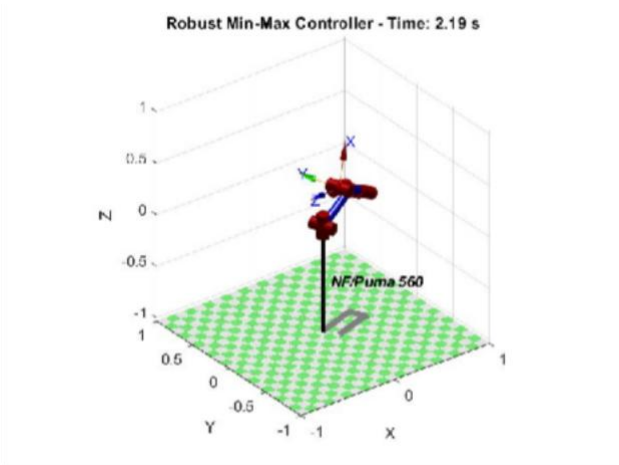

Figure 28-Robust min-max controller simulation

Objective

The goal of this part is to evaluate and compare two control strategies for a PUMA 560 robotic arm performing trajectory tracking:

- Nominal Computed Torque Control (PD + Inverse Dynamics) without knowledge of an added payload.
- Robust Min-Max Control, which compensates for modelling uncertainties like the unknown payload.

The comparison focuses on tracking performance in both joint space and task space, especially under dynamic mismatch (e.g., a 2 kg payload at the end-effector unknown to the controller).

2. Methodology

Robot Model Setup:

- A nominal model without payload and an actual model with a 2 kg payload on Link 6 are defined.
- Friction is neglected in both models using no friction().

Trajectory Generation

- A quintic polynomial trajectory is used for smooth position, velocity, and acceleration profiles from an initial joint configuration q0 to a final configuration qf over 5 seconds.

Controller Implementation

- Nominal Controller: Standard computed torque control using the nominal dynamics and PD feedback.
- Robust Min-Max Controller: Same nominal control law plus a robust term based on the sign and magnitude of the velocity error.

Simulation Execution

- Both controllers simulate the robot's motion using semi-implicit Euler integration.
- Torque commands are clipped to stay within actuator limits for realism.

Error Analysis

- Joint-space errors (e = q_des - q_actual) and task-space errors (transl(fkine(q)) - transl(fkine(q_des))) are computed and plotted for both controllers.

Results & Analysis

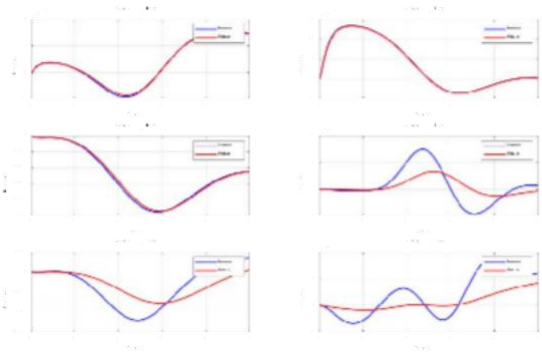

Figure 29-Joint tracking and error for joints(1-6)

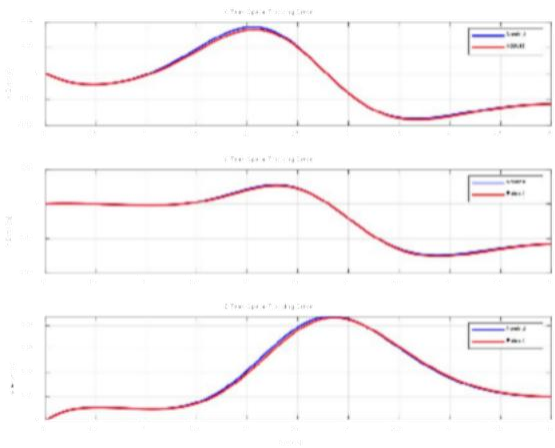

Figure 30-Task-Space Tracking Error

Graph 1: Joint Space Tracking Error – Nominal vs Min-Max Controller

This graph plots the joint tracking error (in radians) over time (0 to 5 seconds) for each of the six joints of the PUMA 560 robot.

Each subplot corresponds to one joint:

- X-axis: Time (seconds)
- Y-axis: Error in joint position (radians)
- Blue curve: Error from the nominal PD + inverse dynamics controller
- Red curve: Error from the robust min-max controller

Joint-by-Joint Interpretation:

- **Joint 1:** Nominal error (blue): Starts with a slight bump (~0.02 rad), settles but shows mild oscillations. Robust error (red): Flatter, quicker settling, indicating better control of rotational base joint despite payload.
- **Joint 2:** Nominal error rises early and shows more fluctuation (~0.06 rad at peak). Robust controller clearly damps the response earlier with smaller steady-state error.
- **Joint 3 :** Nominal error is much larger (peaking above 0.12 rad) – shows that added payload causes under compensation. Robust controller maintains tighter tracking, peaking lower and stabilizing faster.
- **Joint 4, 5, 6:** These joints (wrist) have smaller range of motion but still show: Nominal controller has small but noticeable tracking lag. Min-max error lines are tighter and closer to zero throughout

The robust min-max controller dramatically reduces joint tracking error across all joints, especially where the payload causes significant dynamic mismatch.

This improved tracking helps maintain precise movement and prevents drift or lag, which would otherwise accumulate over multi-joint trajectories.

Graph 2: Task Space Position Error – Nominal vs Min-Max Controller

This graph plots the position error of the end-effector (in meters) over time for the three Cartesian axes (X, Y, Z). This is the task space error, i.e., the difference between the actual and desired position of the robot's hand in 3D space.

Each subplot corresponds to one spatial axis:

- X-axis: Time (seconds)
- Y-axis: Error in position (meters)
- Blue curve: Error from nominal controller
- Red curve: Error from robust min-max controller

Axis-by-Axis Interpretation:

X-Axis Error:

- Nominal controller overshoots initially (~0.03 m) and oscillates more during motion.

- Robust controller follows a smoother trend with less deviation, stabilizing quickly.

Y-Axis Error:

- Nominal controller has visible error and irregular settling.
- Robust controller closely tracks with minimal deviation (below 0.01 m), maintaining precise lateral movement.

Z- Axis Error (vertical motion, heavily affected by payload):

- **Nominal controller** peaks at nearly 0.05 m deviation, unable to compensate for payload's effect on gravity torque.
- **Robust controller** significantly reduces the vertical error, keeping the end-effector aligned even under extra weight.

Takeaway from Graph 2 (Task Space):

- The Z-direction error, which reflects vertical precision and is sensitive to mass changes, shows how the nominal controller fails without payload compensation.
- The robust min-max controller maintains high accuracy in all spatial directions, essential for tasks like assembly, pick-and-place, or surgical applications.

CONCLUSION:

This project implemented and analysed a Computed Torque Control (CTC) strategy for trajectory tracking of a Puma 560 robotic manipulator, combining a PD feedback controller with a feedforward inverse dynamics model to compensate for nonlinearities. The study also evaluated the performance of a robust min-max controller in the presence of model uncertainties, specifically an unknown 2 kg payload at the end-effector. Key findings and conclusions are summarized below:

1. **Effectiveness of Computed Torque Control (CTC):**
   Under nominal conditions with perfect model knowledge, the CTC demonstrated excellent trajectory tracking performance. The integration of inverse dynamics and PD control effectively linearized the system, achieving precise joint-space and task-space tracking with minimal errors. The simulation results confirmed the theoretical advantages of model-based control when dynamic parameters are accurately known.

Impact of payload uncertainty:

The introduction of an unmodeled 2 kg payload revealed limitations in the nominal CTC approach. Joints 2 and 3, which are more sensitive to dynamic changes due to their role in the kinematic chain, exhibited significant tracking errors. This highlighted the sensitivity of model-based controllers to parameter mismatches and unmodeled disturbances.

2. **Robust Min-Max Controller Performance:** The robust min-max controller outperformed the nominal CTC in the presence of payload uncertainty. By incorporating additional compensation for disturbances, the robust controller significantly reduced tracking errors in both joint space and task space. Notably, it mitigated the vertical (Z-axis) error in task space, which was most affected by the payload, demonstrating superior adaptability to dynamic uncertainties.
3. **Trade-offs and Practical Insights:** The results underscored the trade-offs between model-based compensation and robust control. While CTC excels in ideal conditions, its performance degrades with modelling inaccuracies. In contrast, robust controllers like the min-max approach enhance reliability in real-world scenarios where uncertainties are inevitable. This insight is particularly valuable for applications such as industrial automation and collaborative robotics, where payload variations and dynamic changes are common.
4. **Recommendations for Future Work:** Future research could explore adaptive control techniques to further improve performance under varying payloads. Additionally, experimental validation on a physical robot would provide practical insights into the controllers' real-world applicability. Investigating hybrid strategies that combine model-based and learning-based approaches could also be a promising direction.

In conclusion, this study demonstrated the strengths and limitations of computed torque control and robust control strategies for robotic manipulators. The findings emphasize the importance of selecting control methods based on the operational environment, with robust techniques being essential for applications where dynamic uncertainties are prevalent. The insights gained contribute to advancing nonlinear robotic control and optimizing trajectory tracking in practical settings.

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